A COMPLEX SUPPORTED AT CLOSED POINTS OVER A SEPARABLY CLOSED FIELD VANISHES IF AND ONLY IF ALL ITS COHOMOLOGY VANISHES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the trivial direction, the bounded below case using the lowest nonvanishing cohomology sheaf, and the general case through the quasi-compact objects and the identification for U = X.

Editorial note. The label lemma-vanishing-closed-points names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Complexes supported at closed points. Same chapter, the description of complexes whose stalks vanish outside the closed points, and the resulting triviality of their cohomology over the whole scheme. Supplies the identification H^i(X, Q) = H^0(X, H^i(Q)) used in both cases of the proof.
- Cohomology over quasi-compact objects is trivial. Cohomology on Sites, the lemma computing the cohomology of a complex satisfying the support condition as its degree zero part. Supplies the general-case statement that H^q(U, Q) = H^0(U, H^q(Q)) for all quasi-compact objects U.

Source reference. Original reference 0F1F: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 20333–20333. Prerequisite pointer invoked at proof line(s) 20387; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0BKZ: sources/stacks-chow/tags.txt, source lines 14577–14577. Prerequisite pointer invoked at proof line(s) 20394; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Lemma 0.1. *Let K be a separably closed field. Let X be a scheme of finite type over K. Let $Q \in D(X_{étale})$. Assume that $Q_{\overline{x}}$ is nonzero only if x is a closed point of X. Then*

$$Q = 0 \Leftrightarrow H^i(X, Q) = 0 \text{ for all } i$$

Proof. The implication from left to right is trivial. Thus we need to prove the reverse implication.

Assume Q is bounded below; this cases suffices for almost all applications. If Q is not zero, then we can look at the smallest i such that the cohomology sheaf $H^i(Q)$ is nonzero. By Lemma 0F1F we have $H^i(X, Q) = H^0(X, H^i(Q)) \neq 0$ and we conclude.

General case. Let $\mathcal{B} \subset \mathrm{Ob}(X_{étale})$ be the quasi-compact objects. By Lemma 0F1F the assumptions of Cohomology on Sites, Lemma 0BKZ are satisfied. We conclude that $H^q(U, Q) = H^0(U, H^q(Q))$ for all $U \in \mathcal{B}$. In particular, this holds for $U = X$. Thus the conclusion by Lemma 0F1F as Q is zero in $D(X_{étale})$ if and only if $H^q(Q)$ is zero for all q. □